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Supplemental information

Evaluating machine learning models for sepsis prediction: A systematic review of methodologies

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Table S1 Traditional predictive tools and prediction performance, Related to Table 1 and Table3.

Study	Target	Time point	Model	AUROC
1.Delahanty2019	Early prediction of sepsis	24 hours	SOFA	0.90
			SIRS	0.77
			NEWS	0.84
			qSOFA	0.80
			MEWS	0.78
2.Barton2019	Detection and early prediction of sepsis	0 hour	SIRS	0.66
			MEWS	0.61
			SOFA	0.72
			qSOFA	0.60
5.Mao2018	Sepsis detection	0 hour	MEWS	0.76
			SOFA	0.63
			SIRS	0.75
8.Perng2019	Mortality prediction of sepsis	28 days	SIRS	0.59
			qSOFA	0.68
		72 hours	SIRS	0.67
			qSOFA	0.64
11.Kwon2020	Mortality prediction of sepsis	3 days	qSOFA	0.78

Abbreviation: SOFA: Sequential Organ Failure Assessment; SIRS: Systemic Inflammatory Response Syndrome; NEWS: the National Early Warning Score; qSOFA: quick Sequential Organ Failure Assessment; MEWS: Modified Early Warning Score

Table S2 New quality evaluation tool and reevaluation results, Related to Table 2.

Study	Appropriate disease definition	Detailed data preprocessing	Inclusion criteria	Standard data set	Data granularity	Feature Engineering	Sample removing and grouping process	Algorithm	Model evaluation	Prospective clinical validation	Total scores
1.Delahanty2019	1	0	0	1	0	1	1	1	1	0	6
2.Barton2019	1	1	0	1	0	0	0	1	1	0	5
3.Taylro2015	0	1	0	1	0	0	0	1	1	0	4
4.Kam2017	0	1	0	1	0	0	0	1	1	0	4
5.Mao2018	0	1	0	1	0	0	1	0	1	0	4
6.Taneja2017	0	0	0	1	0	1	1	1	1	0	5
7.Saqib2018	0	1	0	1	0	1	0	1	1	0	5
8.Perng2019	0	1	0	1	0	1	0	1	1	0	5
9.Thottakkara2016	0	1	0	1	0	1	0	1	1	0	5
10.Bloch2019	0	0	0	1	1	1	1	1	1	0	6
11.Kwon2020	0	0	0	1	0	0	1	1	1	0	4
12.Nemati2017	1	0	0	1	0	0	0	0	1	0	3
13.Lauritsen2019	0	0	0	1	0	0	1	1	1	0	4
14.Scherpf2019	0	1	0	1	0	0	1	0	1	0	4
15 Hou2020	1	1	1	1	0	1	1	0	1	0	7
16 Kong2020	1	1	1	1	0	0	0	1	1	0	6
17Bedoya 2020	0	0	0	1	0	0	0	0	1	0	2
18 Doorn2021	1	0	0	1	0	0	1	1	1	0	5
19 Li2021	0	1	0	1	0	0	1	0	1	0	4
20 Burdick2020	0	1	0	1	0	0	1	0	1	0	4

21 Qi2021	1	1	1	1	0	0	1	0	1	0	6